

```
//Question no#1
```

```
#include<iostream>
```

```
using namespace std;
```

```
int main(){
```

```
    int roll;
```

```
    float x1, x2, x3, x4, x5, total_marks, percentage;
```

```
    cout << "Enter your roll no: ";
```

```
    cin >> roll;
```

```
    cout << "Enter marks of first subject: ";
```

```
    cin >> x1;
```

```
    cout << "Enter marks of second subject: ";
```

```
    cin >> x2;
```

```
    cout << "Enter marks of third subject: ";
```

```
    cin >> x3;
```

```
    cout << "Enter marks of fourth subject: ";
```

```
    cin >> x4;
```

```
    cout << "Enter marks of fifth subject: ";
```

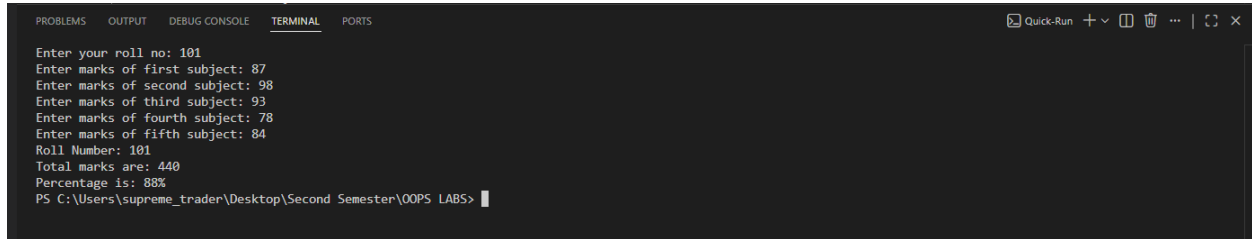
```
    cin >> x5;
```

```
    total_marks = x1 + x2 + x3 + x4 + x5;
```

```
    percentage = (total_marks / 500) * 100;
```

```
    cout << "Roll Number: " << roll << endl;
```

```
cout << "Total marks are: " << total_marks << endl;  
cout << "Percentage is: " << percentage << "%" << endl;  
}
```

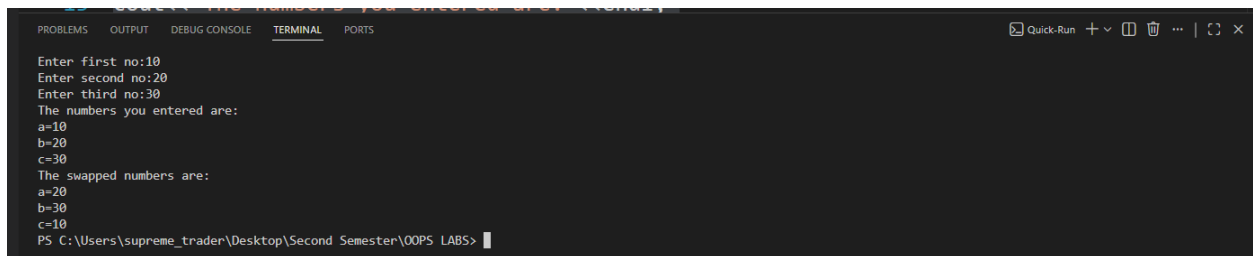
A screenshot of a C++ IDE terminal window. The terminal shows the execution of a program that prompts for a roll number and marks of five subjects. The user has entered a roll number of 101 and marks of 87, 98, 93, 78, and 84 for the five subjects respectively. The program has calculated the total marks as 440 and the percentage as 88%. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the program's output and the command prompt path: PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>.

```
Enter your roll no: 101  
Enter marks of first subject: 87  
Enter marks of second subject: 98  
Enter marks of third subject: 93  
Enter marks of fourth subject: 78  
Enter marks of fifth subject: 84  
Roll Number: 101  
Total marks are: 440  
Percentage is: 88%  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>
```

//Question no#2

```
#include<iostream>  
  
using namespace std;  
  
void swapnumbers(int* a,int* b,int* c);  
  
int main(){  
    int a,b,c;  
  
    cout<<"Enter first no:";  
  
    cin>>a;  
  
    cout<<"Enter second no:";  
  
    cin>>b;  
  
    cout<<"Enter third no:";  
  
    cin>>c;  
  
    cout<<"The numbers you entered are:"<<endl;  
  
    cout<<"a="<<a<<endl;  
  
    cout<<"b="<<b<<endl;  
  
    cout<<"c="<<c<<endl;  
  
    swapnumbers(&a,&b,&c);  
  
    cout<<"The swapped numbers are:"<<endl;
```

```
cout<<"a="<<a<<endl;
cout<<"b="<<b<<endl;
cout<<"c="<<c<<endl;
}
void swapnumbers(int* a,int* b,int* c){
int temp;
temp=*a;
*a=*b;
*b=*c;
*c=temp;
}
```

A screenshot of a C++ IDE terminal window. The terminal shows the following output: Enter first no:10, Enter second no:20, Enter third no:30, The numbers you entered are: a=10, b=20, c=30, The swapped numbers are: a=20, b=30, c=10. The prompt at the bottom is PS C:\Users\supreme_trader\Desktop\Second Semester\00PS LABS>.

```
Enter first no:10
Enter second no:20
Enter third no:30
The numbers you entered are:
a=10
b=20
c=30
The swapped numbers are:
a=20
b=30
c=10
PS C:\Users\supreme_trader\Desktop\Second Semester\00PS LABS>
```

//Question no#3

```
#include<iostream>
using namespace std;
int main(){
    float Fahrenheit,Celsius;
    cout<<"Enter temperature in fahrenheit:"<<endl;
    cin>>Fahrenheit;
    Celsius=(Fahrenheit-32)/1.8;
    cout<<"The temperature in Celsius is:"<<Celsius<<endl;
```

```
}
```

A screenshot of a C++ IDE's terminal window. The window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The terminal shows the following text: "Enter temperature in fahrenheit:", "98.6", "The temperature in Celsius is:37", and a command prompt "PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>". The window also features a "Quick-Run" button and standard window controls (minimize, maximize, close) and a search icon.

//Question no#4

```
#include<iostream>
```

```
using namespace std;
```

```
//Iam using void function with argument
```

```
void addMatrix(int A[3][3], int B[3][3], int sum[3][3]){
```

```
    for (int i=0;i<3;i++){
```

```
        for( int j=0;j<3;j++){
```

```
            sum[i][j]=A[i][j] +B[i][j];
```

```
        }
```

```
    }
```

```
}
```

```
void multiplyMatrix(int A[3][3], int B[3][3], int product[3][3]) {
```

```
    for (int i = 0; i < 3; i++) {
```

```
        for (int j = 0; j < 3; j++) {
```

```
            product[i][j] = 0;
```

```
            for (int k = 0; k < 3; k++) {
```

```
        product[i][j] += A[i][k] * B[k][j];  
    }  
}  
}  
}
```

```
int main ()  
{  
    int A[3][3];  
    int B[3][3];  
    int sum[3][3];  
    int product[3][3];  
    cout<<"\nEnter element of 1st matrix:\n";  
    for (int i=0;i<3;i++){  
        for( int j=0;j<3;j++){  
            cin>>A[i][j];  
        }  
    }  
  
    cout<<"\nEnter element of 2nd matrix:\n";  
    for (int i=0;i<3;i++){  
        for( int j=0;j<3;j++){  
            cin>>B[i][j];  
        }  
    }
```

```
addMatrix(A,B,sum);

cout<<"\nsum of matrices:\n";

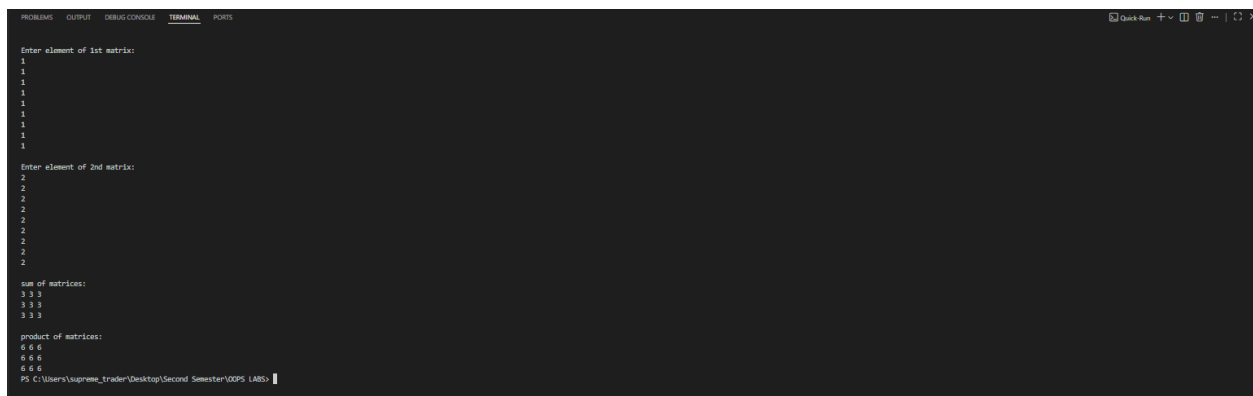
for (int i=0;i<3;i++){
    for( int j=0;j<3;j++){
        cout<<sum[i][j]<<" ";
    }
    cout<< endl;
}

multiplyMatrix(A,B,product);

cout<<"\nproduct of matrices:\n";

for (int i=0;i<3;i++){
    for( int j=0;j<3;j++){
        cout << product[i][j] <<" ";
    }
    cout<< endl;
}

return 0;
}
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Enter element of 1st matrix:
1
1
1
1
1
1
1
1
1
Enter element of 2nd matrix:
2
2
2
2
2
2
2
2
2
sum of matrices:
3 3 3
3 3 3
3 3 3
product of matrices:
6 6 6
6 6 6
6 6 6
PS C:\Users\supreme_trader\Desktop\Second Semester\OODS LABS>
```

```
//Question no#5

#include<iostream>

using namespace std;

float area_of_sphere(float radius);

float volume_of_sphere(float radius);

int main(){

    float radius;

    cout<<"Enter the radius of sphere:";

    cin>>radius;

    cout<<"The area of sphere is:"<<area_of_sphere(radius)<<endl;

    cout<<"The volume of sphere is:"<<volume_of_sphere(radius)<<endl;

}

float area_of_sphere(float radius){

    float area;

    area=4*3.142*radius*radius;

    return area;

}

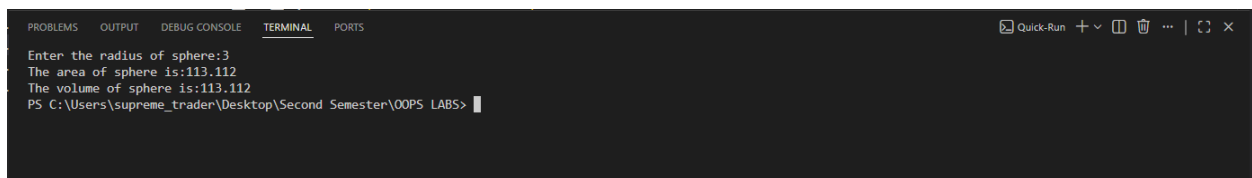
float volume_of_sphere(float radius){

    float volume;

    volume=(4.0/3.0)*3.142*radius*radius*radius;

    return volume;

}
```

A screenshot of a C++ IDE terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the output of a program. The program prompts the user to enter the radius of a sphere, which is 3. It then calculates and displays the area of the sphere as 113.112 and the volume of the sphere as 113.112. The command prompt shows the current directory as C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Enter the radius of sphere:3
The area of sphere is:113.112
The volume of sphere is:113.112
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>
```

//Question no#6

```
#include <iostream>
```

```
using namespace std;
```

```
double savingsWithdraw(double amount, double balance)
```

```
{
```

```
    double charge = amount * 0.02;
```

```
    double tax = 0;
```

```
    if (amount > 50000)
```

```
    {
```

```
        tax = amount * 0.05;
```

```
    }
```

```
    balance = balance - amount - charge - tax;
```

```
    cout << "Account Type: Savings" << endl;
```

```
    cout << "Charge (2%): " << charge << endl;
```

```
    cout << "Tax: " << tax << endl;
```

```
    return balance;
```

```
}
```

```
double currentWithdraw(double amount, double balance)
```

```
{
```

```
    double charge = 100;
```

```
    double tax = 0;
```

```
    if (amount > 50000)
```

```
    {
```

```
        tax = amount * 0.05;
```

```
    }
```



```
balance = balance - amount - charge - tax;

cout << "Account Type: Current" << endl;
cout << "Charge: " << charge << endl;
cout << "Tax: " << tax << endl;

return balance;
}
int main()
{
    char accountType;
    int accountNumber;
    double withdrawAmount;
    double balance = 200000;

    cout << "Enter account type (S for Savings, C for Current): ";
    cin >> accountType;
    cout << "Enter account number: ";
    cin >> accountNumber;
    cout << "Enter withdrawal amount: ";
    cin >> withdrawAmount;
    if (withdrawAmount > 100000)
    {
        cout << "You cannot withdraw more than 100000 at a time." << endl;
        return 0;
    }
}
```

```
if (accountType == 'S' || accountType == 's')
{
    balance = savingsWithdraw(withdrawAmount, balance);
}
else if (accountType == 'C' || accountType == 'c')
{
    balance =
        currentWithdraw(withdrawAmount, balance);
}
else
{
    cout << "Invalid account type!" << endl;

    return 0;
}

cout << "Remaining Balance: " << balance << endl;

return 0;
}
```



The screenshot shows a terminal window with the following text:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Enter account type (S for Savings, C for Current): S
Enter account number: 12345
Enter withdrawal amount: 60000
Account Type: Savings
Charge (2%): 1200
Tax: 3000
Remaining Balance: 135800
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS>
```

The terminal window has a dark background and a light-colored text. The top bar shows tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. The output shows the program's execution flow, including user input and calculated values.